

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the
Interstate Transmission System

Docket No. RM26-4-000

**COMMENTS OF THE CALIFORNIA DEPARTMENT OF WATER
RESOURCES STATE WATER PROJECT**

The California Department of Water Resources State Water Project (“DWR”) appreciates the opportunity to comment on the October 23, 2025, Advance Notice of Proposed Rulemaking (“ANOPR”),¹ which requests comments on “potential reforms to ensure the timely and orderly interconnection of large loads to the transmission system.”² DWR appreciates the Federal Energy Regulatory Commission’s (the “Commission”) efforts to address large load interconnections and provides the below comments on certain Principles for Reform included in the ANOPR.

Specifically, DWR comments on Principles Seven, Eight, and Thirteen:

- Principle Seven: The Commission should not establish expedited study treatment for large loads that agree to be curtailable, due to the potential for discriminatory access and increased costs to ratepayers. Instead, the Commission should adopt market-based approaches to encourage load—both new and existing—to be curtailable, which will mitigate undue discrimination concerns and more fully unlock the benefits of curtailable load.
- Principle Eight: DWR supports holding large loads responsible for financing 100% of the network upgrade costs they are assigned, which will protect other customers from bearing costs they did not cause. This assignment of costs should be paired with an appropriate crediting mechanism that keeps other wholesale transmission customers whole.

¹ Secretary of Energy, Advanced Notice of Proposed Rulemaking (released Oct. 23, 2025), eLibrary No. 20251027-4001 (“ANOPR”).

² ANOPR P 1 (footnote omitted).

- Principle Thirteen. As part of a plan to implement any large load interconnection reforms, the Commission must respect regional differences and include the independent entity variation for RTOs/ISOs.

I. DESCRIPTION OF DWR

The California Department of Water Resources is an agency of the State of California, headquartered in Sacramento. It is responsible for monitoring, conserving, and developing California's water resources, providing flood protection to ensure public safety, and preventing property damage related to water resources. A primary responsibility of DWR is the construction, operation, and maintenance of the State Water Project ("SWP").

The SWP is the largest state-owned, multi-purpose water project in the country. Its operations are critical to the resources and economy of the state. The SWP's system spans nearly the entire state, from Upper Feather River reservoirs and Lake Oroville in Northern California to Pyramid, Castaic, Silverwood and Perris reservoirs in Southern California. The SWP delivers an average of 2.6 million acre-feet of water per year to 29 public agency water contractors throughout California. Approximately 40% of the deliveries are used to irrigate approximately 750,000 acres of farmland. The rest of the deliveries serve the water needs of more than 27 million Californians.

In order to serve these needs, the SWP's water conveyance system includes 29 water storage facilities, approximately 700 miles of aqueducts and pipe lines, 21 pumping plants, three pumping-generating plants and five hydroelectric power plants. The SWP's power generating sources have capacity of over 1,595 megawatts, and generate 3 to 5 million megawatt-hours of energy per year. The SWP's pumping facilities have a capacity of approximately 2,576 megawatts, and consume 6 to 9.5 million megawatt-hours of energy per year. In addition to its hydroelectric facilities, the SWP receives

power from long-term contracts. Furthermore, the SWP manages its power operation through self-generation, load management including demand response, purchase and sales transactions with other entities, and participation in the California Independent System Operator Corporation (“CAISO”) power markets.

II. COMMUNICATIONS

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III. COMMENTS

- A. **Principle Seven. To incentivize load to be curtailable, load should be enabled to participate in the regional markets to the maximum extent possible.**

DWR appreciates the Commission’s focus on curtailable load and agrees that there are benefits to curtailable load that should be encouraged through appropriate

means. Rather than offering expedited interconnection on a preferential basis, the Commission should consider other market incentives to encourage these resources. Preferential access to the queue raises the potential for game playing—many generators could conform their projects to include new, curtailable load, thus crowding the queue for other generators and new loads. New loads that cannot accept curtailment under all conditions might have more trouble interconnecting and face higher costs. Instead, the Commission should direct transmission providers to enable load to participate in their respective markets to the maximum extent possible. Maximized market participation, including through load-side bidding and demand response, is preferable to a jump-the-line approach as the market will deliver more effective incentives to all new and installed curtailable load, which will result in more available system capacity than a one-time incentive only available to new load interconnections.

Expedited study treatment is a less effective means of unlocking the benefits of curtailable load, because it is a one-time incentive that would only apply to new large load interconnection requests. An expedited study process approach entirely ignores existing load that could be curtailable (or more curtailable)—leaving valuable curtailable capacity on the table.

Instead, the Commission should promote better utilization of curtailable load through market solutions that enable curtailable load to participate in markets and that appropriately compensate it for the value it provides. Such market-based approaches do not present the same undue discrimination concerns as a one-off, expedited study for certain load interconnections, as they link the incentive (e.g., market payments) to the particular benefit provided by any load capable of curtailment.

Moreover, the incentives for large loads to curtail that are provided through well-designed market-based solutions will update over time as either the amount of curtailability or its value changes. This flexibility and precision are particularly important given the changes in technology relative to large loads, particularly data centers. Data center load is proving to be increasingly flexible as their operational focus changes from cloud services to artificial intelligence.³ New data center technology can delay certain energy-intensive tasks to times when generator curtailments are high or when other demand is low.⁴ They can also shift their cooling operations to optimal timeframes.⁵ But these shifts have economic impacts on large load—shifting cooling can shorten the life of data center components, for example.⁶ A market solution could be a win-win for large load and ratepayers—it provides an economic solution to potentially uneconomic conduct by the large load, while decreasing the overall load burden on existing systems, which could require less transmission, distribution, and generation build-out (and thus less impact on ratepayers).

DWR is the largest single load participating in the CAISO markets to the extent allowed. CAISO's markets do not currently allow real-time load bidding, but CAISO is

³ Tyler H. Norris et al., *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems* at 11, Nicholas Institute for Energy, Environment & Sustainability (2025), <https://nicholasinstitute.duke.edu/sites/default/files/publications/rethinking-load-growth.pdf> (“Unlike the many real-time processing demands typical of conventional data center workloads, such as cloud services and enterprise applications, the training of neural networks that power large language models and other machine learning algorithms is deferrable. This flexibility in timing . . . allows for strategic scheduling of training as well as other delay-tolerant tasks.”).

⁴ *Id.*

⁵ *Id.* at 12 (“[O]perators can increase cooling during midday when solar energy is abundant and reduce cooling during peak evening demand.”).

⁶ *Id.* (stating that shifting cooling “may be perceived as uneconomic due to potential impacts on performance”).

exploring this functionality through its ongoing stakeholder initiative process.⁷ DWR appreciates CAISO's ongoing development of real-time load bidding because it provides system benefits for little additional cost to ratepayers. CAISO's Department of Market Monitoring ("DMM") supports real-time load bidding functionality because it can provide such benefits as "increase[d] market reliability and efficiency" and "lower[] real-time system costs."⁸ Reducing load in real-time allows "existing on-line capacity to meet the needs of the system without additional supply. The additional supply could then contribute to reserves for tight conditions, and as a result lower real-time system costs because the market would not be dispatching the next marginal unit."⁹

Unlocking the curtailability of existing loads requires no upgrades or increased ratepayer costs—it simply requires new market functionality.¹⁰ If CAISO implements real-time load bidding, DWR can provide approximately 2,600 MW of available curtailable load, because the State Water Project's water pumping operations are curtailable in response to market signals. DWR can increase its load during times of low demand and high curtailment and decrease its load during times of high demand, thereby providing the system benefits described by CAISO's DMM.

⁷ See CAISO, *Initiative: Demand and Distributed Energy Market Integration*, CAISO Stakeholder Center, <https://stakeholdercenter.caiso.com/StakeholderInitiatives/Demand-Distributed-Energy-Market-Integration> (last visited Nov. 18, 2025).

⁸ DMM, *Comments on Demand and Distributed Energy Market Integration Working Group* at 2, 5 (Nov. 6, 2025), <https://stakeholdercenter.caiso.com/Common/DownloadFile/18566d0b-731c-410d-a7fa-d1b47391f855>.

⁹ DMM, *Comments on Demand and Distributed Energy Market Integration Working Group Discussion Paper* at 3 (July 11, 2025), <https://stakeholdercenter.caiso.com/Common/DownloadFile/74a0fcb7-b9be-4540-8020-872ec8691f36>.

¹⁰ DWR notes that in CAISO's case, it simply requires *restored* market functionality. Certain types of load were able to bid into the real-time market prior to CAISO's 2008 market redesign. CAISO has yet to restore the functionality despite its increasing value as generation curtailments increase within CAISO.

Enabling load to participate in the markets to the extent possible is consistent with past Commission statements and practice.¹¹ The Commission’s support for demand-side bidding is more important than ever as load increases across the nation—the Commission should reiterate that support and order transmission providers to enable load to participate in the markets to the extent possible.

B. Principle Eight. Wholesale customers should be held harmless for new load interconnections.

New large interconnections should finance 100% of the network upgrades they are assigned through the interconnection study process. New large load interconnections, such as those for data centers, are creating challenges due to their unique size, pace of growth, and distinctive characteristics.

While the forecasts for these large loads are massive, they are also uncertain and unpredictable. In September 2025, then-Chairman Rosner issued a letter to RTOs/ISOs focused on load growth from large data centers and advanced manufacturing, recognizing that “we cannot efficiently plan the electric generation and transmission needed to serve new customers if we don’t forecast how much energy they will need as accurately as possible.”¹² The responses confirmed the unique challenges of forecasting these large

¹¹ See, e.g., *Regional Transmission Organizations*, Order No. 2000, 89 FERC ¶ 61,285, FERC Stats. & Regs. at 31,217 (1999), *order on reh’g*, Order No. 2000–A, 90 FERC ¶ 61,201 (2000) (“Demand-side bidding is desirable to the extent it is technically feasible, because without it, demand response decreases and market power is easier to exercise. The availability of price responsive demand also reduces price volatility in the markets.”) (footnote omitted); *Wholesale Competition in Regions with Organized Electric Markets*, Order No. 719, 125 FERC ¶ 61,071, P 18 (2008), *corrected*, 126 FERC ¶ 61,261, *on reh’g*, Order No. 719-A, 128 FERC ¶ 61,059, *on reh’g*, Order No. 719-B, 129 FERC ¶ 61,252 (2009). FERC’s goal of fully integrating demand-side resources into organized markets should not “be delayed until every barrier to demand response, whether retail or wholesale, technological or regulatory, is identified and addressed.” *Wholesale Competition in Regions with Organized Electric Markets*, 122 FERC ¶ 61,167, P 109 (2008).

¹² David Rosner, *Chairman Rosner’s Letter to the RTOs/ISOs on Large Load Forecasting*, FERC (Sep. 18, 2025), <https://www.ferc.gov/news-events/news/chairman-rosners-letter-rtosisos-large-load-forecasting>.

loads. In response, CAISO explained that it has “developed a ‘large loads’ team across the CAISO to analyze large load issues.”¹³

The uncertainty and unpredictability of large load forecasts presents risks of unjust, unreasonable, and unduly discriminatory transmission rates. If massive transmission investments are made to serve these loads, yet the expected demand does not materialize, other customers may be left holding the bag. These wholesale customers have already incurred the costs necessary to accommodate their normal load growth (and in some instances have paid for available capacity that may be used up by new large loads). To prevent existing wholesale customers from bearing disproportionate costs for new load interconnections, the Commission should adopt an approach, consistent with Principle Eight, that holds large loads such as data centers responsible for financing the network upgrades triggered by their interconnections.

This assignment of costs should be paired with an appropriate crediting mechanism. This crediting mechanism might vary from region-to-region, but it must be designed to keep wholesale customers whole and disincentivize speculative load that will impose costs on wholesale customers. Appropriate crediting requirements for this purpose will need to look different than crediting mechanisms used for generator interconnections. Under the CAISO large generator interconnection procedures, for example, “the Interconnection Customer shall be entitled to a repayment for the cost of Network Upgrades” to be paid “within five (5) years of the Commercial Operation

¹³ Letter from Elliot Mainzer, President and Chief Executive Officer of CAISO, to FERC Chairman David Rosner re Large Load Forecasting at 2 (Oct. 15, 2025), <https://www.ferc.gov/media/caiso-response-chairman-david-rosners-09192025-letter-re-large-load-forecasting-americas>.

Date.”¹⁴ Such a guaranteed crediting mechanism should not be adopted for large load interconnections because these credits are not tied to particular levels of usage. In addition, the existing guaranteed crediting mechanism is not designed to address the unique cost impositions that large load interconnections could have on other customers. Instead, a crediting system that is designed to address the problems of speculative large loads should include requirements that ensure a significant portion of the requested load shows up over a sustained period of time, in order to disincentivize over-reservation of transmission capacity and thus over-building of transmission to serve a load that is smaller than expected.

A system that is designed to hold existing wholesale customers harmless will properly balance allowing new load access to the transmission system while not allowing speculative load interconnection requests to increase costs to existing customers. The Commission should aim to strike this balance by assigning network upgrades to the responsible new load interconnection, paired with an appropriate crediting mechanism.

C. Principle Thirteen. The Final Rule should respect regional differences.

Priorities and situations across the country with respect to large load integration differ, as do the existing interconnection procedures and tariffs into which large load interconnection procedures would have to be integrated. The Final Rule should adopt principles that guide the RTOs in their treatment of large load interconnections—but each RTO should be given the latitude to adopt a program that works best for its region.

¹⁴ CAISO Tariff, App. U § 3.4.3.

Consistent with the Commission’s approach to generator interconnection procedures and agreements,¹⁵ the Commission should evaluate RTO/ISO implementation of any final rule in light of the independent entity variation for RTOs/ISOs. Just as in the context of the Commission’s standard generator interconnection rules, this flexibility is important to recognize efforts already underway to address these issues and to “encourage transmission providers to continue to innovate to remedy their identified interconnection queue management issues.”¹⁶

IV. CONCLUSION

DWR appreciates the opportunity to provide comments on the important topics raised in the ANOPR and urges the Commission to (1) encourage market-based approaches to incentivizing new and existing load to be curtailable instead of providing expedited studies for new large loads that agree to be curtailable; (2) hold new large loads responsible for financing 100% of the network upgrade costs they are assigned through the interconnection study process, paired with an appropriate crediting mechanism that keeps other wholesale transmission customers whole; and (3) respect regional differences by evaluating any RTO/ISO implementation of a final rule using the independent entity variation standard.

¹⁵ See, e.g., *Improvements to Generator Interconnection Procedures and Agreements*, Order No. 2023, 184 FERC ¶ 61,054, P 10 (2023), *on reh’g*, 185 FERC ¶ 61,063 (2023), *on reh’g*, Order No. 2023-A, 186 FERC ¶ 61,199, *errata notice*, 188 FERC ¶ 61,134 (2024) (“We note that the compliance obligations that result from this final rule will be evaluated in light of the independent entity variation standard for regional transmission organizations (RTO) and independent system operators (ISO) . . .”).

¹⁶ *Id.*

Respectfully submitted,

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